Project Context:

In this project, I am a Supply Chain Analyst at Boeing, based in Kankakee, Illinois (not real location). I am pulling a dataset full of vendor order information, and I am trying to find the vendors that have the highest amount of "delayed" orders. The parameters of a delayed order were set in my Python code, and they are as follows: Estimated Delivery Days: 7 to 14 days. I created an "Actual Delivery Days" to create some randomization when it came to the delivery's arrival times, and I made that range 7 to 21 days, here were the results.

Abc supply_chain_summary.csv Vendor	supply_chain_summary.csv Location	# supply_chain_summary.csv Total Orders	# supply_chain_summary.csv Delayed Orders	# supply_chain_summary.csv Delay Rate
Adam's Collection	Milwaukee, WI	27	20	0.740741
Bob's Burgers	Indianapolis, IN	20	17	0.850000
John's Products	Chicago, IL	22	14	0.636364
Sam's Storage	Detroit, MI	19	14	0.736842
Tim's Inc	St. Louis, MO	12	10	0.833333

I'll get into the interpretation of these results later, but I want to describe my mental process when I was creating this dataset, and how and why I wrote the queries that I did to pull this data.

Coding Methodology:

(Disclaimer: I used Gemini as a Teacher and a guide to help me with unknown syntax and libraries)

I used Python to create the supply chain database (supply_chain.db) that my SQL queries would eventually collect and sort data from. Since I didn't really know of any online .db files I could pull from, I decided that I wanted to create as realistic of an order data base as I could. You can see how I constructed my databases in my 'supply_chain.sql' tab in GitHub, but the main columns that I made Python create and randomize data for were order_id (total_orders), order_status, estimated_date, and actual_date. I learned a few tricks on how the 'Random' library works when creating data for a .db, and I really wanted to carefully create this portion of my code, so that the data that was created would be as realistic as possible. For my SQL queries, I wanted to create a basic query that would pull all Delayed Orders, and then from there, I wanted to refine and refine until I got each vendor, their location, their total orders, and total delayed orders. I made vendor_id the foreign key in my 'orders' table, and my primary key in my vendors table. And to pull the count of total orders and delayed orders, I simply used the COUNT aggregate to tally the total orders (used order_id), and used a CASE statement to differ the 'delayed' orders from the total, using "order_status = 'Delayed'". The delay rate was not something I added inside my SQL tables or queries, but instead was added in my Python code.

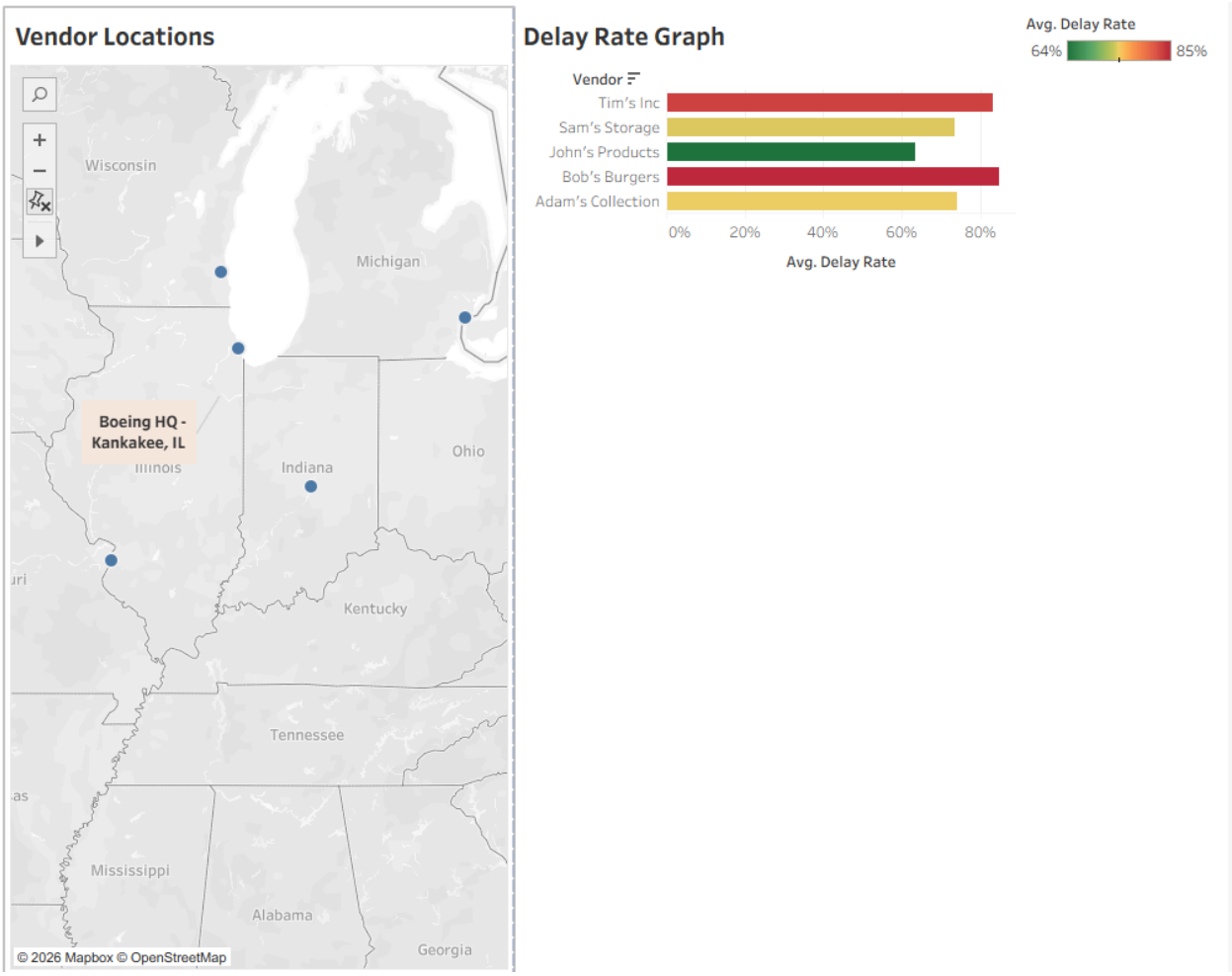
I had to create a CSV file for Tableau (couldn't read .db's unfortunately), and it helped create a bar graph for comparing delay rates between vendors.

Analytical Assumption and Interpretation:
I ran into a problem: I created the code to create random data for total units ordered, shipping costs, location of vendors, and stopped to consider: Wouldn't delayed orders have a higher correlation to those 3 factors if the amount of units were higher, shipping costs were higher, and the distance from the vendor to my company (Boeing) was further away than the others were? I almost went into the code to create probability factors that would give higher units in an order a higher chance of being 'delayed' for example, but I am glad I didn't. Because the data isn't what makes these orders delayed, it's what happens at the vendor sites, the supply and demand of the companies needed parts, and a multitude of other factors that father those symptoms (symptoms that I thought were the delta of the delayed orders). That's when I realized having these numbers being totally randomized is what makes this such a realistic scenario for any kind of Data/Supply/Business Analyst: you have to interpret what those numbers are saying, and not make assumptions when you don't have the full picture. And for this fake scenario at a Boeing site in Kankakee, IL, I don't have the full picture. I don't have that enterprise, C-Suite information to know why Bob's Burgers out of Indianapolis had the highest Delay Rate out of the 5 vendors, but for the purposes of this project, I do know why that and Tim's Inc share the highest Delay Rate: Bob's procurement process is the worst of the group. They are always having equipment issues, and they seem to never have the right materials to even fully create the parts that need to be shipped to Boeing. And Tim's Inc honestly got the short end of the stick. They are the smallest vendor out of the 5, but they got 12 orders this month, 5 more than normal (I'm creating the 'normal' for the purpose of the project), so the amount of drivers and trucks could not keep up with the orders that came in, and that also affected their assembly line work: during peak hours of order fulfillment, the line was not properly ready to handle that high influx of orders, leading to delays.

Disclaimer: the other vendors all have obnoxiously high delay rates, but just analyzing the two highest for simplicity's sake.

Delay Rate Graph and Geographic Map:

I uploaded the CSV with my order data into Tableau to create two simple visualizations: a bar graph, and a geographical map:



It's not the prettiest thing, but it allows me to compare the vendor delay rate and their respective location to our Boeing site in Kankakee.

Conclusion:

When analyzing either supply chain data, or sales conversion data, the job of the analyst is to clean, aggregate, sort, and interpret that data to their leaders to help improve business decision making. Hopefully with the information on why Bob's Burgers and Tim's Inc have their problems, that Boeing Corporate can address those procurement/labor shortage issues head on, to improve those delay rates.